

How fIAIr scores a match

Every time an animal is reported, fIAIr compares it to every listing on the opposite side (lost against found, and vice versa). It assigns points based on resemblance, proximity and dates, then keeps only the credible leads, ranked from strongest to weakest. fIAIr is a **filter that sorts**, not an oracle : a human always confirms in the end.

1. Automatic exclusions

Before any points are counted, a lead is discarded if :

- the **species** differs (a cat cannot match a dog) ;
- the animal was **found more than 2 days before** the disappearance date (chronologically impossible) ;
- the **colours are clearly different** (no shared shade) and the face is not visually similar. This is what stops, for instance, a black cat from matching a grey cat.

2. The points system

CRITERION	CONDITION	POINTS
Coat colour	Identical	+35
	Shared shade ("black and white" vs "black")	+15
	Different, but face very similar (rescued by the AI)	-12
Breed	Exact specific breed (Siamese, Labrador, Bernese Mountain Dog...)	+25
	Same generic category (domestic shorthair/longhair, mixed)	+5
Visual resemblance AI face fingerprint	Very strong	+28
	Strong	+16
	Moderate	+6
Eye colour	Identical	+12
Proximity	See the distance scale below	0 to +20
Dates	Found within ~2 weeks after the disappearance	+8
	Found up to ~6 weeks after	+4

3. The distance scale

≤ 5 km Very close · +20	≤ 25 km Close · +12	≤ 75 km Same area · +6	> 75 km No points · 0
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Distance is measured as the crow flies between where the animal went missing and where it was found. Beyond 75 km it adds nothing : the lead must then rest entirely on physical resemblance.

4. The probability level

Once all points are added up, fIAIr rates the lead on this scale :

Not shown under 35 points	Moderate match 35 to 54 points	Strong match 55 points and up
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Below 35 points, a lead is deemed too weak and is never shown. fIAIr displays at most the 6 best leads, from strongest to weakest.

5. Two concrete examples

False positive, discarded

Croque-mou, a **black** cat. Candidate : a **grey** cat, 38 km away, same domestic shorthair, same yellow eyes.

- Clearly different colours
- No visual face resemblance
- Generic breed (barely counts)

→ **Discarded automatically**

Strong match

A **black-and-white** cat is lost. Found candidate : a **black-and-white** cat, 3 km away, found 2 days later, very similar face.

- Identical colour : +35
- Very close (≤ 5 km) : +20
- Found soon after : +8
- Very similar face : +28

→ **91 points · Strong**

fIAIr proposes, humans decide. The system surfaces the most plausible leads to save you time ; the final confirmation always rests with the family and the person who found the animal. Scoring subject to adjustments as real data comes in. · animALERTE.ca